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GP Demand and Demographic Change:

Forecasting Major Disease Prevalence to 2043

by

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Executive Summary

General Practice (GP) is the cornerstone of the Scottish healthcare system, providing first contact, primary care, and coordination with secondary services. Rising demand—driven by population aging, multimorbidity, and workforce pressures—makes it essential to anticipate how GP caseloads will evolve in the coming decades.

Previous forecasts of disease burden relied on outdated 2020-based population projections and lacked post-COVID prevalence data, limiting their accuracy and policy relevance. This study integrates two new sources: 2022-based post-Brexit population projections from the Office for National Statistics (ONS) and practice-level disease prevalence data published by Public Health Scotland in 2025. Using Bayesian hierarchical regression, it produces novel forecasts of disease burden and GP demand in Scotland to 2043, disaggregated to health boards, health and social care partnerships, council areas, and data zones.

Key findings:

- Demographic drivers: Age and sex are dominant predictors. Cardiovascular and metabolic conditions rise steeply with age, while depression remains concentrated in younger cohorts.
- Substantial growth: By 2043, hypertension is projected to approach one million patients nationally. Across conditions, case counts rise by 10–23%, closely tracking population growth and ageing.
- Local variation: Practices account for much greater unexplained variation than health boards or partnerships, indicating that local characteristics matter. Boards with older or more deprived populations will face disproportionate growth in burden.
- Workload effects: Higher patient-to-GP ratios are associated with greater prevalence, particularly for depression.

Although projections are constrained by fragmented and inconsistent data, short prevalence series, and simplifying assumptions about future epidemiology, they nonetheless provide internally consistent, empirically grounded scenarios. These results indicate not exact patient counts but the scale, direction, and distribution of future demand.

Implications for policy:

- Provides a ready framework for incorporating GP demand into the Whole System Model under development by Public Health Scotland.
- Highlights where capacity pressures will intensify most—cardiovascular care, diabetes management, and mental health—and which regions are likely to face disproportionate strain.
- Underscores the need for targeted workforce planning, resource allocation, and investment in practice resilience, especially in areas serving older or more deprived populations.

Recommendations for next steps:

1. Extend prevalence reporting to cover more years and ensure complete practice coverage.
2. Develop direct subnational population projections, rather than relying on proportional scaling.
3. Incorporate richer measures of GP supply (e.g. working hours, skill mix, practice organisation).
4. Introduce dynamic epidemiological trends to move beyond purely demographic extrapolation.

Despite limitations, this study establishes a scalable, transparent, and data-driven framework for forecasting GP demand. It provides timely, policy-relevant evidence to support national planning and ensure that Scottish general practice remains resilient and responsive to changing population needs over the next two decades.

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- Given the sources of all pictures, data etc. that are not my own
- Not made any use of the report(s) or essay(s) of any other student(s) either past or present
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1 Introduction

General Practice (GP) is the first point of contact for most people engaging with the Scottish healthcare system, providing both primary care and coordination with hospitals, community clinics, and medical specialists. Rising demand for these services—driven by population ageing, multimorbidity, and other pressures—has made understanding patterns of GP use, and projecting how they will evolve, critical for effective national planning and resource allocation.

In 2022, the Scottish Public Health Observatory—a collaboration between Public Health Scotland (PHS), the Glasgow Centre for Population Health, and various partner organizations—published a study projecting the future burden of critical diseases to 2043.[13] However, that work relied on 2020-based population projections. These estimates were produced only months after Brexit and at the onset of the COVID-19 pandemic, and there is subsequently justification to believe they were biased or quickly outdated.[11]

In 2023, the Office for National Statistics released revised 2022-based population projections reflecting post-Brexit demographic shifts.[12] More recently, in 2025, GP-practice-level prevalence data for major diseases became available for the first time.[15] Taken together, these developments present a timely opportunity to generate more accurate and policy-relevant forecasts of healthcare demand.

This project builds on that opportunity. Using Bayesian regression models (via `brms`) fitted to practice populations, disease prevalence, and service activity, it projects the burden of major diseases across Scotland's regions to 2043 at multiple scales: health boards, health and social care partnerships, council areas, and datazones. By integrating updated demographic forecasts with new prevalence data, this analysis refines national disease burden estimates and grounds them in the operational realities of GP services.[13, 12]

The study has three aims: (1) to describe variation in GP populations and disease prevalence across practices and regions; (2) to identify the demographic, socioeconomic, and workforce factors driving this variation; and (3) to project future disease counts and GP demand under post-Brexit population scenarios. In doing so, it provides an evidence base for incorporating GP services into the Whole System Model currently being developed by PHS, supporting workforce planning, resource allocation, and policy decisions to ensure that Scotland's healthcare system is better prepared to meet changing population needs.

2 Data

2.1 Initialization

This study draws primarily upon datasets published through Public Health Scotland's open data platform (opendata.nhs.scot). These sources collectively describe the demographic composition of GP practice populations, the size of the GP workforce, and the geographic and socioeconomic context in which practices operate.[9, 7, 8, 4, 5]

Patient-level demand is measured through the *GP Practice Population Demographics* dataset, which reports quarterly counts of registered patients by practice, sex, and five-year age bracket from 2014 to 2025.[9] From these records, I construct a time series of patient counts by age and sex at the practice level.

Workforce supply is derived from the *General Practitioner Contact Details* dataset, which identifies the individual GPs employed at each practice between 2018 and 2025.[7] These records are aggregated into annual counts of GPs per practice, providing a longitudinal measure of staffing capacity.

Geographic accessibility is estimated using the *GP Practice Contact Details and List Sizes* dataset, which reports addresses and list sizes from 2015 to 2025.[8] Postcodes are converted into geographic coordinates, from which inter-practice distances are calculated to approximate the availability of alternative providers within a catchment.

All three datasets consistently link practices to one of Scotland's 6,796 data zones, the standard small-area geography for official statistics. Data zones nest within health and social care partnerships (HSCPs), which in turn aggregate to Scotland's 14 health boards. For this study, projections are harmonized across data zones, HSCPs, health boards, and council areas. A supplementary dataset, *Data Zone 2011*, provides the mapping necessary to align practices with these geographies.[5]

Local socioeconomic context is introduced through the Scottish Index of Multiple Deprivation (*SIMD 2020v2*), which assigns each data zone both a deprivation rank and a decile across the nation.[4] These variables allow relative affluence and disadvantage to be incorporated directly into the analysis of GP demand.

Disease prevalence is obtained from the *Disease Prevalence in General Practice* report, published by PHS in July 2025.[15] This dataset provides counts and rates of 19 conditions across Scottish GP practices between 2022 and 2025. For modeling, I focus on the eight most prevalent conditions nationally—diabetes, depression, hypertension, asthma, cancer, coronary heart disease, chronic kidney disease, and stroke/TIA—as they represent the bulk of the disease burden in primary care. A previous prevalence report covering 2018–2022, released in 2022 and updated in 2024, is also referenced in the code for comparative purposes, though it is not central to the present analysis.[14]

Population projections are drawn from two sources published by the Office for National Statistics (ONS): the 2022-based national projections (to 2047) and the 2018-based council-area projections (to 2043).[12, 10] The 2022-based series serves as the baseline, as it reflects post-Brexit demographic conditions. However, these projections are published only at the national level. To generate subnational estimates, I first extract annual population shares by council area from the 2018-based series and apply these proportions to the 2022-based totals to approximate council-area populations through 2043. These are then proportionally disaggregated to data zones and HSCPs using 2022 population distributions (sourced from PHS Population Estimates), yielding harmonized projections across all geographies of interest.[6] The final population dataset excludes years 2044–2047, as no subnational scaling is possible beyond 2043.

The final working dataset therefore includes:

- the number of patients with a given disease per practice, age bracket, sex, and year;
- the number of total patients at a given practice by age bracket, sex, and year;
- the number of GPs at a given practice by year;
- the data zone, HSCP, health board, and council a given practice is in;
- the SIMD rank and decile of each data zone (and average for higher regions); and
- the distance between each practice and the next closest one.

2.2 Exploratory Data Analysis

To motivate the specification of the regression models, I first examine the principal sources of variation in the GP and disease prevalence data. These analyses establish the empirical foundations for including practice workload, demographic composition, socioeconomic deprivation, geographic hierarchy, and accessibility as covariates.

The distribution of patients per GP, shown in Figure 1, is right-skewed. Most practices cluster tightly just above 1,000 patients per GP, with the majority lying between 500 and 2,000. A small number of outliers extend far beyond this range, with ratios exceeding 5,000 and in rare cases approaching 10,000. This skewness underscores workload as a central source of practice-level variation and the right skew motivates considering a log transformation.

Disease prevalence exhibits systematic demographic patterns. Figure 2a shows prevalence by age group for the eight selected conditions. For most diseases—cancer, chronic kidney disease, coronary heart disease, hypertension, and stroke/TIA—rates rise steeply with age, while asthma and depression are concentrated in younger cohorts. Figure 2b highlights sex differentials: coronary heart disease, diabetes, and stroke are more common among men, while depression and asthma are more common among women. These gradients confirm the necessity of including age, sex, and their interaction in the models.

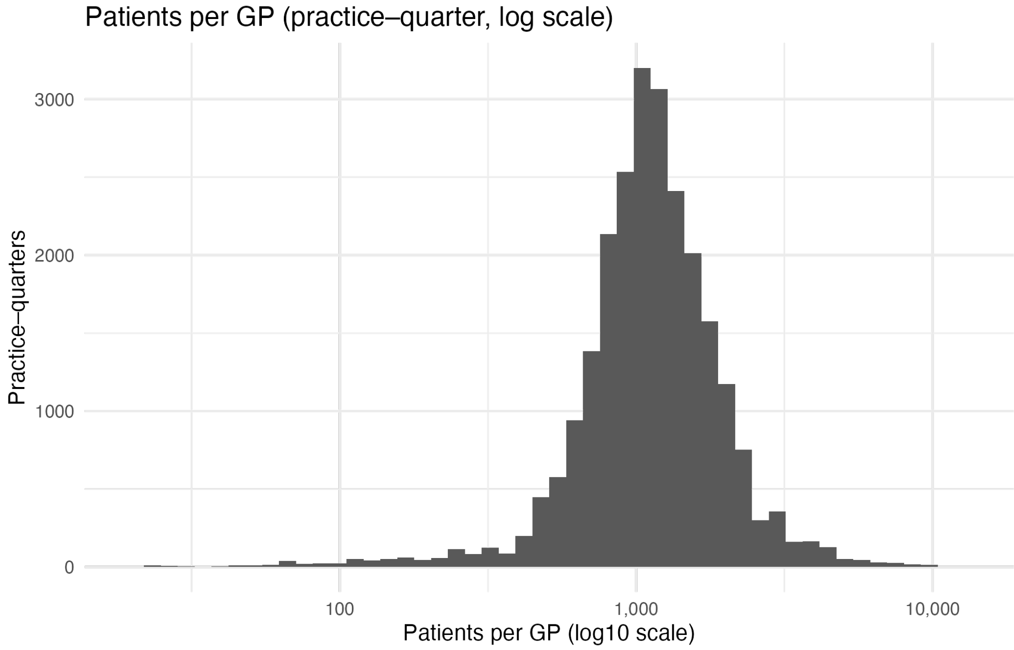

Figure 1: Patient counts per GP aggregated by quarter, on log scale to approach a Normal distribution.

Socioeconomic context also plays a role. Figure 3 plots practice-level prevalence against SIMD decile. Most conditions show higher prevalence in more deprived communities, with cancer the notable exception, which tends to show little or even slightly higher prevalence in less deprived areas.

Geographic context adds further heterogeneity. Figure 4 displays prevalence distributions by Health Board. Substantial between-area variation is evident, with practices in some boards showing consistently higher or lower prevalence across multiple conditions.

Finally, practice accessibility is proxied by the distance to the nearest alternative GP. Disease prevalence shows weak but consistent associations with distance (Figure 5). While not a dominant effect, these patterns support considering distance as an additive covariate.

3 Models

3.1 Bayesian Hierarchical Regression

To model disease prevalence at the practice level, I employ Bayesian hierarchical regression using the `brms` package in R. This framework allows flexible specification of likelihood families, link functions, and multilevel structures while providing full Bayesian inference through Hamiltonian Monte Carlo in Stan. Compared with frequentist generalized linear mixed models, the Bayesian approach yields full posterior distributions for all parameters, supports principled uncertainty quantification, and permits explicit prior specification. This is particularly important in a health services context, where variation arises at multiple geographic and organizational levels, and where uncertainty itself is central to policy interpretation.

3.2 Model Structure

Three complementary model formulations are pursued, each fit in both extended and reduced forms and compared with leave-one-out cross-validation (LOO). This two-stage design balances the need for disease-specific tailoring with the need to keep the overarching framework consistent. The three families are as follows.

3.2.1 Binomial Model

$$y_i \sim \text{Binomial}(n_i, \theta_i) \tag{3.1}$$

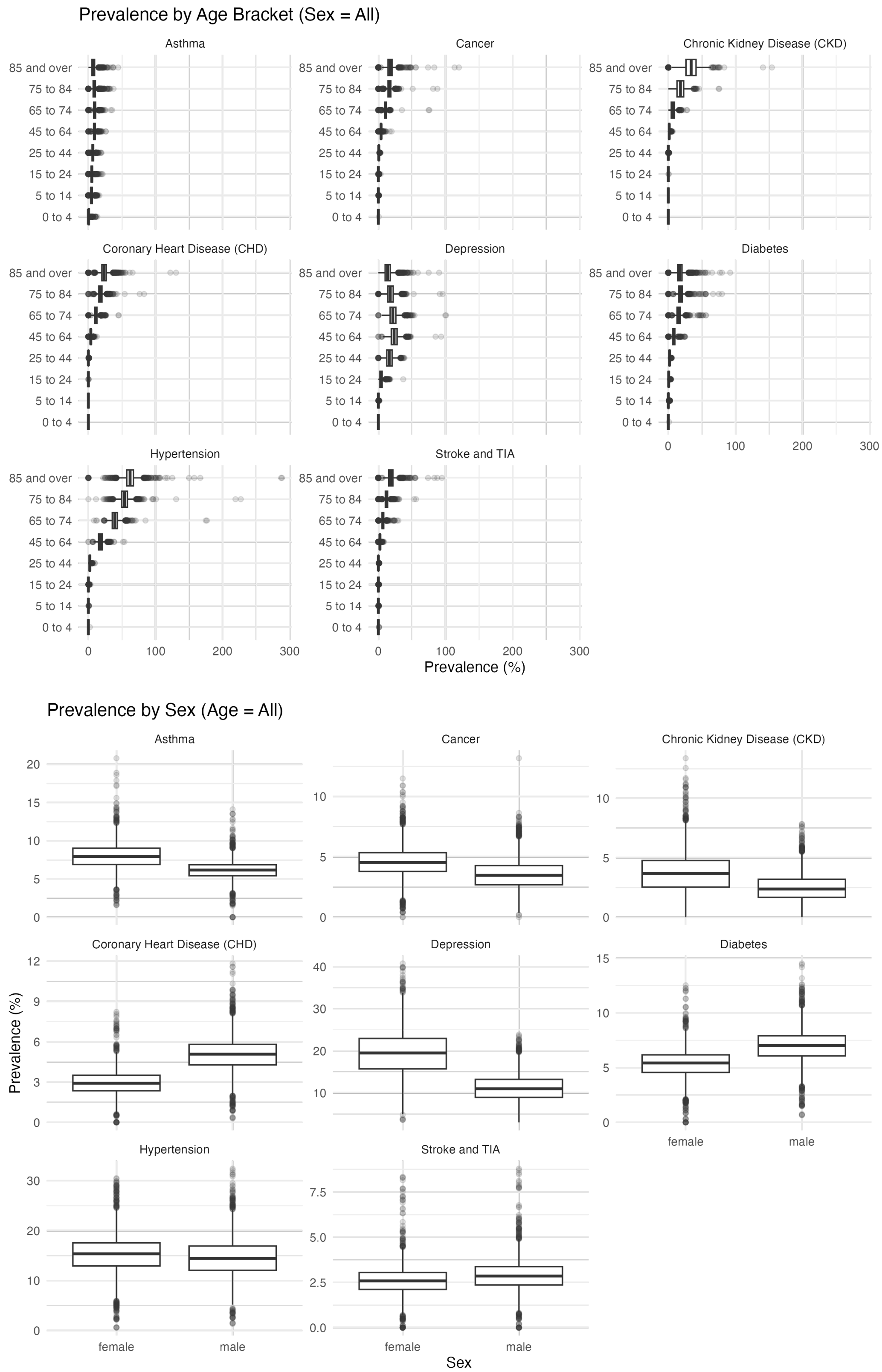

Figure 2: Disease prevalence by age and sex.

where y_i is the number of cases, n_i is the number of patients, and θ_i is the latent prevalence probability for stratum i.

The binomial model is the natural starting point: in principle, each patient is a Bernoulli trial with probability θ_i. However, denominators n_i are not direct patient counts but rounded, quarterly averages from the *GP Practice Population Demographics* dataset, while numerators y_i are aggregated

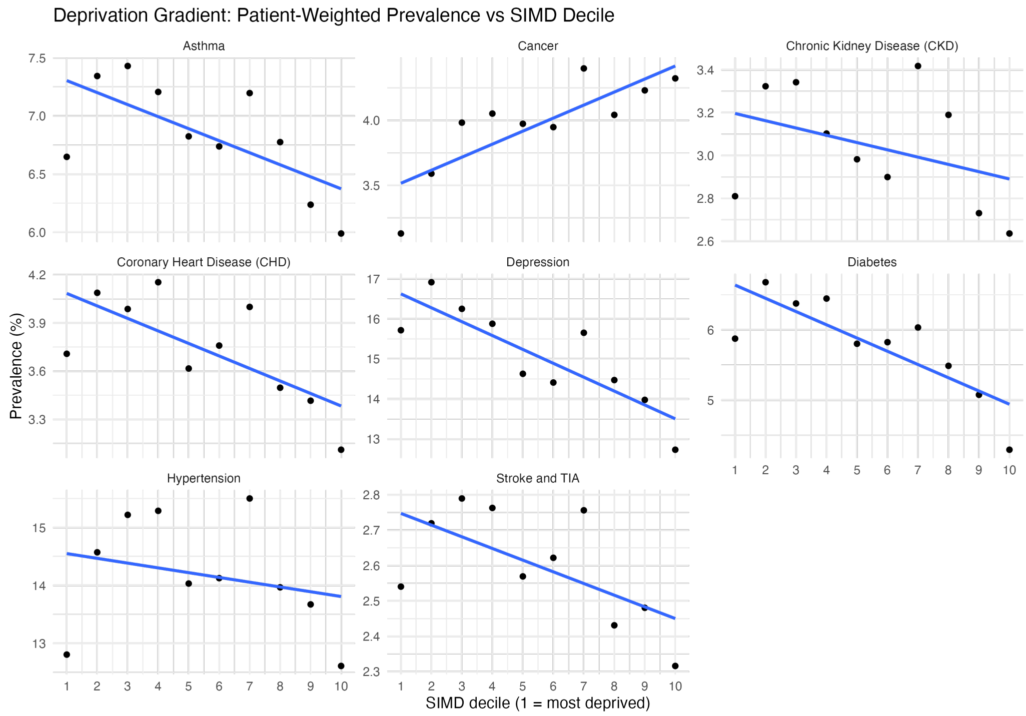

Figure 3: SIMD rank per practice.

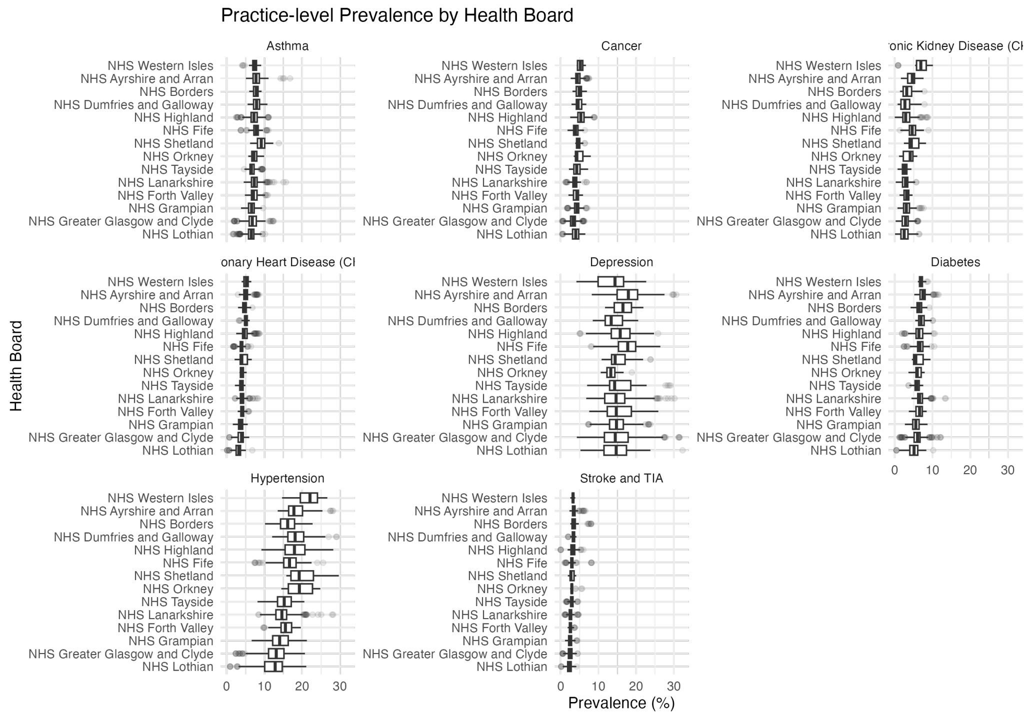

Figure 4: Disease prevalence by health board.

from the *Disease Prevalence in General Practice* report. This mismatch means the binomial variance,

$$\mathrm{Var}(y_i) = n_i\theta_i(1-\theta_i), \tag{3.2}$$

may overstate precision, since the numerator and denominator are not measured on the same underlying population.

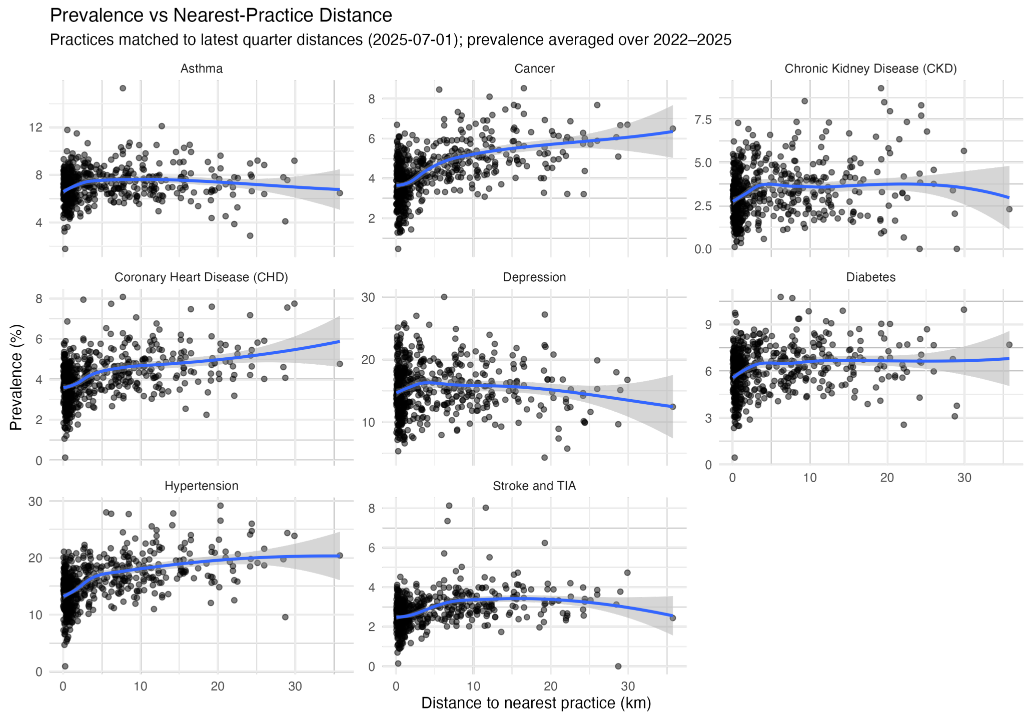

Figure 5: Disease prevalence by distance to next-closest practice (i.e. rurality).

3.2.2 Beta Model

$$z_i \sim \mathrm{Beta}(\mu_i \phi, (1-\mu_i)\phi) \tag{3.3}$$

where $z_i = \frac{\text{Disease Count}}{\text{Patient Count}}$ is the observed prevalence expressed as a proportion in $(0, 1)$, μ_i is the mean prevalence, and ϕ is a precision parameter.

The beta model discards explicit patient counts and instead models proportions directly. This allows variance to be governed by ϕ rather than mechanically tied to the mean and denominator, making it better suited to settings with imprecise denominators or where small denominators would otherwise dominate uncertainty. The drawback is that strata of very different sizes are treated on equal footing.

3.2.3 Zero-One Inflated Beta (ZOIB) Model

A compromise is provided by the ZOIB formulation, which augments the beta model with explicit mass at 0 and 1:

$$z_i \sim \begin{cases} 0 & \text{with probability } \pi_0, \\ 1 & \text{with probability } \pi_1, \\ \mathrm{Beta}(\mu_i\phi,\ (1-\mu_i)\phi) & \text{with probability } 1-\pi_0-\pi_1. \end{cases} \tag{3.4}$$

In this context, ZOIB guards against artifacts introduced when extremely small denominators or rounding produce apparent 0% or 100% prevalence strata. It thus bridges the binomial's patient-count logic and the beta's flexible dispersion, providing robustness in edge cases.

3.2.4 Common Structure

All three likelihoods share a common predictor structure. For each observation i, defined by practice p, HSCP h, health board b, age bracket a, sex s, and year t:

$$\begin{aligned}\operatorname{logit}(\theta \text{ or } \mu) \;&= \beta_0 + \beta_1 \log\!\left(\tfrac{\text{Patient Count}}{\text{GP Count}}\right) \\ &\quad + \beta_2\,\text{Age} + \beta_3\,\text{Sex} + \beta_4\,(\text{Age} \times \text{Sex}) \\ &\quad + \mathbb{I}(\text{ext}) \times \big(\beta_5\,\text{SIMD} + \beta_6\,\text{Distance}\big) \\ &\quad + u_{\text{practice}} + h_{\text{HSCP}} + b_{\text{HB}}.\end{aligned} \tag{3.5}$$

Here, θ_i is the latent prevalence probability in the binomial model and μ_i is the mean prevalence in the beta and ZOIB models. Random intercepts u_{practice}, h_{HSCP}, and b_{HB} account for clustering within practices, HSCPs, and health boards.

The indicator $\mathbb{I}(\text{ext})$ determines whether deprivation (SIMD decile) and distance to nearest practice are included. For each disease and family, both extended and reduced forms were first fit to a practice-level subsample, and predictive performance compared with LOO. The superior form (or the reduced form if near indistinguishable) was then carried forward as the final model. The ZOIB model was treated as a variant of the beta family for this purpose.

3.3 Priors

To regularize estimation without overwhelming the data, weakly informative priors centered at plausible scales were adopted.

For fixed effects, regression coefficients followed Normal priors:

$$\beta_k \sim N(0,1),$$

which encode the belief that covariates are unlikely to shift log-odds prevalence by more than a few units, while permitting wide exploration if supported by the data. The intercept was given a similar $N(0,1)$ prior, reflecting lack of strong prior knowledge about baseline prevalence once predictors are centered.

Random intercepts were modeled as

$$u_{\text{practice}} \sim N(0,\sigma_p), \quad h_{\text{HSCP}} \sim N(0,\sigma_h), \quad b_{\text{HB}} \sim N(0,\sigma_b),$$

with group-level standard deviations following half-Student-t distributions:

$$\sigma \sim t^{+}(3,0,2.5).$$

This prior reflects the recommendation of Gelman et al. (2008), who show that heavy-tailed half-t distributions with moderate scale provide a weakly informative yet stable default for hierarchical variance parameters.

For the beta family, an additional prior was placed on the precision parameter:

$$\phi \sim \text{Gamma}(2, 0.1),$$

favoring moderate dispersion while remaining weak enough to allow wide variation when supported by the data.

3.4 Model Comparison and Selection

Extended and reduced specifications within each family were compared using approximate LOO-CV on a subsample of practices. Where the extended form yielded a clear improvement in expected log predictive density (ELPD), it was retained; otherwise, the reduced form was preferred. For the beta family, the ZOIB variant was evaluated in parallel but treated as an extension of the beta model.

Table 1 reports the LOO results across all eight conditions. Within each family, the model with highest ELPD is shown in bold. For instance, the reduced beta model was favored for asthma, cancer, chronic kidney disease, and diabetes, while the extended form was preferred for coronary heart disease, depression, hypertension, and stroke/TIA. In the binomial family, reduced models were selected for asthma, cancer, chronic kidney disease, depression, and diabetes, with extended forms preferred for coronary heart disease, hypertension, and stroke/TIA.

Following this within-family step, the best-performing beta and binomial models for each disease were refit on the full dataset, with chain length and warmup doubled to ensure stable inference. This yielded a final pair of models (beta and binomial) per condition for subsequent evaluation.

Table 1: LOO model comparison by disease. Values are expected log predictive density (ELPD). Best model within each family in bold.

	Beta			Binomial	
Disease	Reduced	Expanded	ZOIB	Reduced	Expanded
Asthma	**6504.108**	6504.023	6497.896	**-9227.691**	-9227.918
Cancer	**11209.06**	11208.77	11202.07	**-5798.160**	-5798.456
Chronic Kidney Disease	**12601.32**	12599.90	12594.50	**-4312.392**	-4313.283
Coronary Heart Disease	11764.93	**11767.68**	11761.29	-5628.478	**-5627.964**
Depression	7655.838	**7655.909**	7649.412	**-15865.68**	-15866.38
Diabetes	**9651.105**	9650.756	9644.776	**-7053.511**	-7053.672
Hypertension	9079.498	**9083.013**	9074.984	-8249.402	**-8249.317**
Stroke/TIA	11779.64	**11781.74**	11775.97	**-4790.426**	-4790.689

3.5 Posterior Predictive Checks

All full models demonstrated satisfactory convergence, with $\hat{R} < 1.01$ and high effective sample sizes across parameters. Posterior predictive checks (PPCs) were used to evaluate how well the models captured key distributional features of the observed data. Across conditions, both families reproduced mean prevalence reasonably well, but a consistent divergence appeared in the variance structure. The binomial models tended to underestimate empirical variability, particularly in strata with larger denominators, a direct consequence of the variance being mechanically tied to $n_i\theta_i(1-\theta_i)$.

In contrast, the beta models, by introducing the additional precision parameter ϕ, flexibly accommodated overdispersion and provided a closer match to the observed spread of prevalence values. As illustrated in Figure 6 for depression, the beta family captured both the mean structure and the empirical standard deviations more faithfully. The same pattern held across all eight conditions, giving consistent evidence in favor of the beta specification.

For this reason, although both families were carried forward through the model selection stage, subsequent interpretation emphasizes the beta results, as they yield the more reliable uncertainty quantification and better reflect the data-generating process in practice.

4 Results

The fitted models provide a coherent picture of the demographic, socioeconomic, and practice-level factors shaping variation in disease prevalence across Scottish general practices.

4.1 Fixed Effects

Posterior estimates of fixed effects (Figure 7) highlight the central role of age, which emerged as the strongest predictor in all conditions: log-odds effects ranged from 1.3 for stroke/TIA to 2.4 for hypertension. GP workload, proxied by the log ratio of patients per GP, was also consistently positive, with particularly strong associations for depression ($\beta = 1.0$) and asthma ($\beta = 0.72$). Sex differences were condition-specific: men had higher prevalence of diabetes and coronary heart disease, while

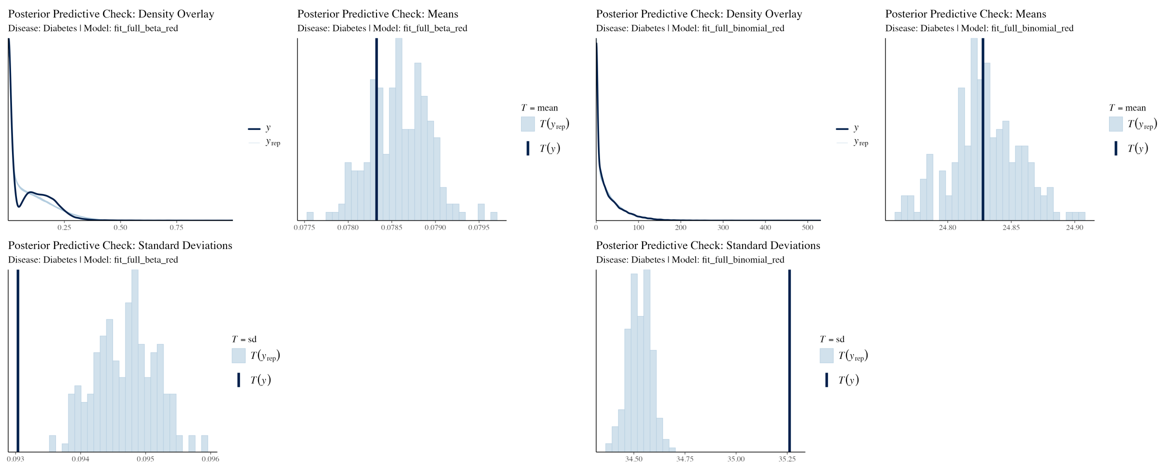

Figure 6: Posterior predictive checks for full models of depression prevalence. Density overlays, sample means, and sample standard deviations are shown for observed data (y) and replicated draws (y^{rep}). While both families capture mean prevalence, the beta model more accurately reflects the observed variability. This pattern was consistent across all eight conditions examined.

women were at higher risk of depression and asthma. Deprivation gradients were modest overall, though detectable for depression and hypertension, and distance to the nearest practice had small negative effects for coronary heart disease and stroke/TIA.

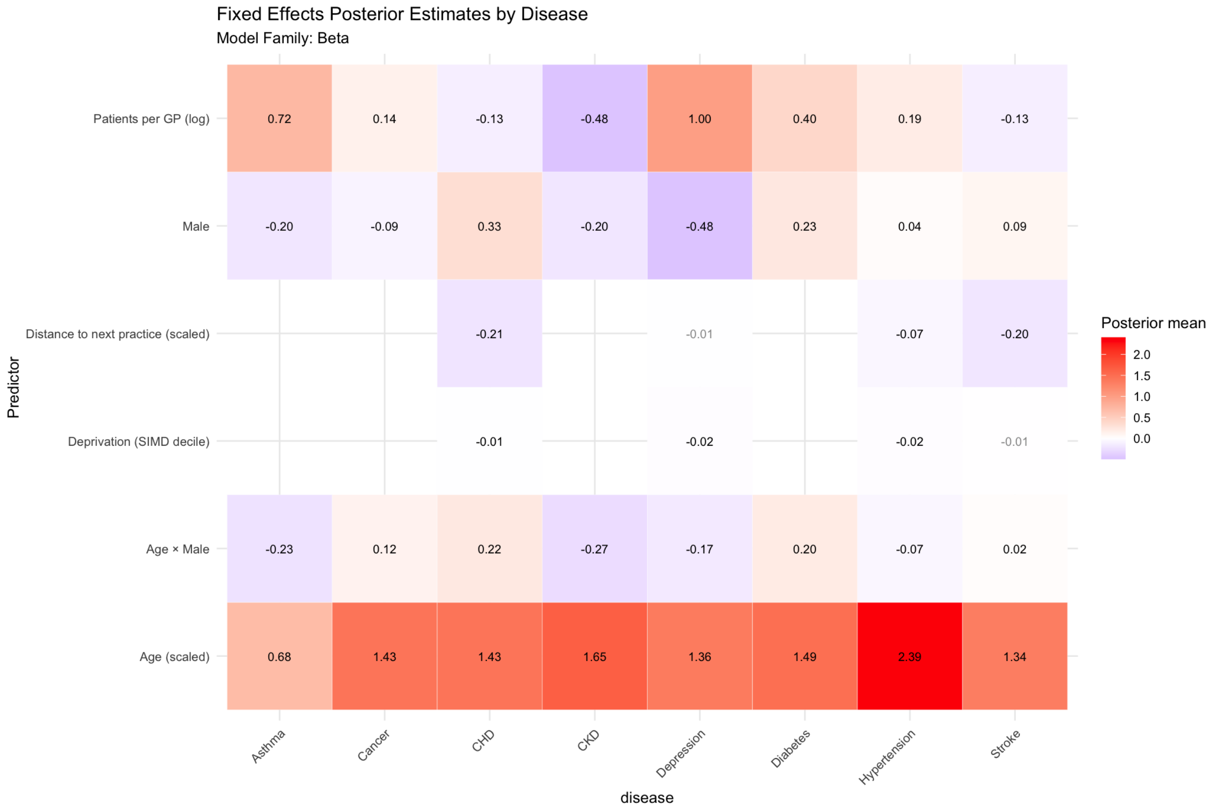

Figure 7: Posterior means of fixed effects coefficients across diseases (Beta family models). Red indicates positive association, blue negative.

4.2 Random Effects

Random intercepts revealed substantial heterogeneity between practices, markedly exceeding variation at HSCP or Health Board levels (Figure 8). CKD and depression showed the largest practice-level variance (posterior sd ≈ 0.66 and 0.53, respectively). This pattern, consistent across diseases, indicates that local practice characteristics capture much of the unexplained burden, while regional clustering is weaker but non-negligible. Such variation reinforces the need for projections to be resolved at fine geographic scales, rather than only at the national level.

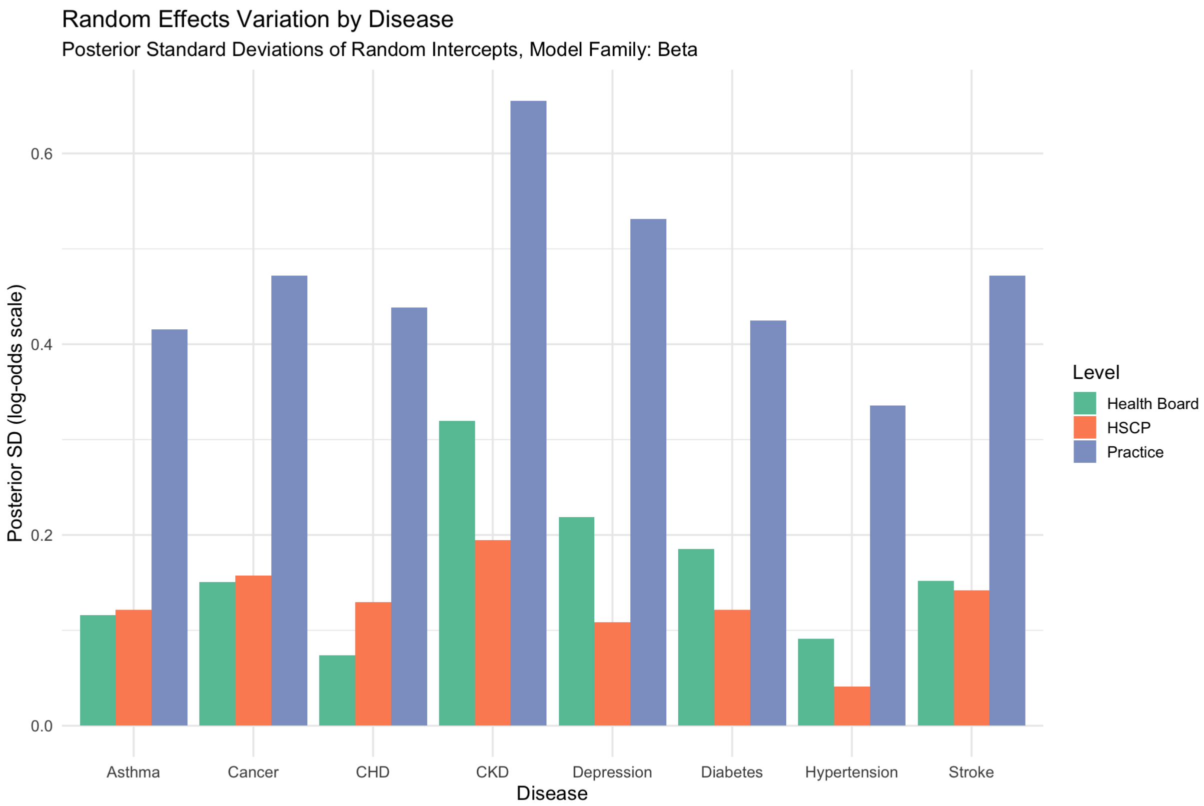

Figure 8: Posterior standard deviations of random intercepts by disease and geographic level (Beta family models).

4.3 Condition-Specific Findings

Condition-specific summaries (see Appendix for full posterior tables) illustrate these patterns in more detail.

- **Diabetes:** Age was the dominant driver ($\beta = 1.49$), with slightly higher prevalence among men ($\beta = 0.23$). Workload was positively associated, and practice-level variation moderate (sd $= 0.42$).
- **Depression:** Showed the clearest socioeconomic gradient (higher prevalence in more deprived areas, $\beta = -0.02$ per decile). Women had substantially higher prevalence ($\beta = -0.48$ for men relative to women). Workload was strongly positive ($\beta = 1.0$), and practice heterogeneity high (sd $= 0.53$).
- **Hypertension:** Displayed the steepest age gradient ($\beta = 2.39$), with weaker effects from workload ($\beta = 0.19$) and deprivation ($\beta = -0.02$). Practice-level variance was moderate (sd $= 0.34$).
- **Asthma:** Concentrated among younger groups ($\beta = 0.68$) and more common in women ($\beta = -0.20$ for men). Workload effects were again positive ($\beta = 0.72$).
- **Cancer:** Primarily age-driven ($\beta = 1.43$), with minimal socioeconomic or distance effects.
- **CHD:** Age ($\beta = 1.43$) and male sex ($\beta = 0.33$) were the strongest predictors, with modest distance and deprivation effects.

- **CKD:** Strong age gradient ($\beta = 1.65$), modestly lower prevalence among men ($\beta = -0.20$), and the largest practice-level heterogeneity (sd = 0.66).
- **Stroke/TIA:** Age again dominated ($\beta = 1.34$), with small positive male effect ($\beta = 0.09$).

5 Projections

5.1 Methodology

The regression results in Section 4 identified age and sex as the most important predictors of prevalence across all conditions. This is advantageous for our analysis, since official population forecasts are published precisely in terms of sex and age structure. By combining fitted regression results with general population projections, we are therefore able to simulate future prevalence and disease counts under various population scenarios.

Projections were generated using the final Beta models selected in 3. For each disease, posterior draws were extended through 2043 for all practice-aggregated strate and then aggregated upwards to councils, HSCPs, health boards, and national totals. Expected prevalence rates were multiplied by projected population counts in each stratum to yield disease case estimates.

Population projections were taken from the 2022-based series published by the Office for National Statistics [12], harmonized to Scottish council areas using the 2018-based series, and further disaggregated to HSCPs and data zones according to the method described in Section 2[10, 13]. To account for demographic uncertainty, three projection variants were utilized:

- **Principal:** baseline scenario, reflecting ONS's best-estimate assumptions on fertility, mortality, and migration;
- **Low:** lower-growth scenario, combining lower fertility and net migration with higher mortality;
- **High:** higher-growth scenario, combining higher fertility and net migration with lower mortality.

These variants provide a plausible envelope for the projected number of patients and are interpreted here as demographic uncertainty bounds around the principal scenario.

Table 2 summarises projected national counts under the principal scenario, with low and high variants shown for 2043. Across all eight conditions, the results indicate substantial increases in patient numbers, driven primarily by population ageing and growth.

Table 2: Projected national disease counts, Scotland 2023–2043 (principal scenario, with low and high population variants as bounds).

Disease	2023 Count	2043 Count	2043 Low	2043 High	Pop. Change
Asthma	278,053	307,239	291,627	319,634	+10.5%
Cancer	226,674	274,367	265,170	281,260	+21.0%
CKD	207,014	252,709	244,498	258,840	+22.1%
CHD	245,319	298,681	288,670	306,176	+21.7%
Depression	502,352	587,882	566,383	604,248	+17.0%
Diabetes	289,539	352,093	340,517	360,739	+21.6%
Hypertension	798,009	983,457	955,551	1,003,966	+23.3%
Stroke/TIA	183,473	220,154	212,380	226,025	+20.0%

The largest absolute increases are projected for hypertension, which rises to nearly one million patients by 2043, and depression, which approaches 600,000. Relative increases fall between 10% and 23% across conditions, closely tracking overall population growth. Because prevalence rates are modelled as stable conditional on demographics, most of the observed increases reflect changes in population size and age structure rather than shifts in underlying risk.

6 Discussion

This study set out to provide updated forecasts of disease burden and GP demand in Scotland by combining newly available practice-level prevalence data with revised post-Brexit population projections. The analysis had three aims: (1) to describe variation in GP populations and disease prevalence across practices and regions; (2) to identify the demographic, socioeconomic, and workforce factors shaping this variation; and (3) to project future disease counts under alternative demographic scenarios. Each of these aims has been addressed, though with important caveats.

6.1 Objectives

The first aim was met through descriptive and hierarchical modelling, which confirmed demographic structure as the main driver of prevalence. Cardiovascular and metabolic conditions rose steeply with age, while asthma and depression were higher in younger groups. Sex differences matched established patterns, and deprivation/workload effects varied by condition. Most unexplained variation lay at the practice level. The second aim was addressed via regression analysis. Age and sex were the strongest predictors; deprivation had modest overall effects but was important for depression and hypertension. Workload (patients per GP) was positively linked to prevalence, especially for depression and asthma. Thus, while demographics dominate, socioeconomic and organisational factors matter too.

The third aim was met through national and subnational projections to 2043, combining fitted regression structures with ONS population forecasts. Table 2 summarised national totals, Figure 9 juxtaposes observed counts (2022–2025) with projected trajectories through 2043, and Figure 10 presents projected trajectories by disease and health board. Across all eight conditions, patient numbers are projected to rise substantially. Hypertension alone is expected to approach one million patients nationally. Relative growth rates track population change closely, indicating that demographic ageing, rather than changes in underlying risk, is the primary driver.

6.2 Implications

The projections carry several implications for GP services and planning. Cardiovascular and metabolic conditions (hypertension, diabetes, and CHD) are projected to contribute the largest increases, underscoring the importance of long-term management strategies in primary care. Depression, already highly prevalent, remains the single largest condition in absolute terms, highlighting the continuing scale of mental health needs. Subnational disaggregation further shows that boards and partnerships with older or more deprived populations will face greater increases, suggesting that resource allocation cannot be uniform but must reflect local population structures.

The modelling framework developed here—linking practice-level prevalence data to updated demographic forecasts—provides a foundation for incorporating GP demand into the Whole System Model being developed by Public Health Scotland. In that context, the projections are best understood as order-of-magnitude scenarios that highlight where and how demand will grow, rather than as precise headcounts.

6.3 Limitations

Several limitations constrain the precision and interpretation of these findings:

1. **Fragmented data sources.** The study integrates datasets from multiple providers, each with slightly different coverage. Practices occasionally fail to report, geographies are periodically revised, and denominators are constructed inconsistently across series. These mismatches introduce noise and potential bias.

2. **Short and incomplete prevalence series.** Practice-level prevalence is available only from 2022 and remains incomplete for some practices. Non-reporting practices were excluded or indirectly imputed, risking bias if their populations differ systematically. With only three years of prevalence data, underlying epidemiological trends cannot be reliably assessed.

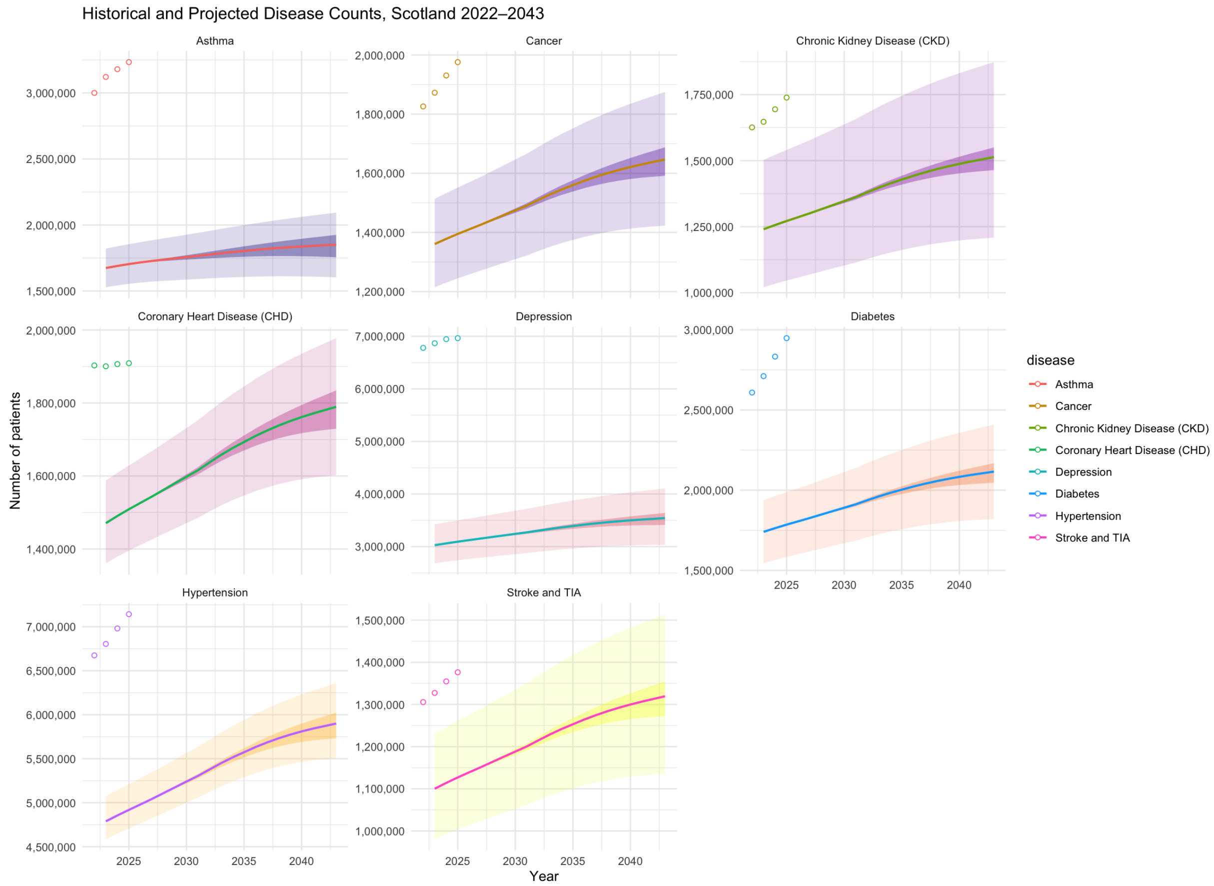

Figure 9: Historical and projected disease counts, Scotland 2022–2043. Points represent historical data; lines represent projections under the ONS principal scenario; inner ribbons show low–high population variants; outer ribbons show posterior predictive 95% credible intervals.

3. **Population projections and scaling.** Subnational populations were derived by applying 2018-based council shares to 2022-based national totals and disaggregating to HSCPs and data zones. This assumes spatial distributions remain constant, which is unrealistic and may distort projections for areas with high migration or distinctive fertility and mortality patterns.

4. **Quarterly averaging of patient counts.** Patient denominators were constructed by averaging quarterly snapshots. While this smooths volatility, it also obscures within-year changes and may misalign with prevalence counts, leading to potential bias in binomial-type models.

5. **Mixing of rate and count sources.** In some instances, prevalence estimates and population denominators were drawn from different sources. Despite harmonisation, definitional mismatches (e.g. treatment of temporary residents or multimorbidity) risk over- or under-estimation. This helps explain why some historical points fall outside projected uncertainty bounds.

6. **Simplified modelling of GP supply.** GP workforce was proxied by headcount, without adjustment for hours worked, skill mix, or reliance on advanced practitioners. The patient-to-GP ratio therefore only partially reflects true practice capacity.

7. **Assumption of stable epidemiology.** Projections assume that conditional prevalence by age and sex remains constant. Any changes in risk factors, treatments, or health behaviours will alter actual prevalence, potentially diverging from forecasts.

These limitations mean the projections should not be interpreted as precise forecasts of absolute numbers. Instead, they provide internally consistent scenarios that highlight demographic and geographic pressures on GP services. Their primary value lies in identifying broad patterns of demand, not exact totals.

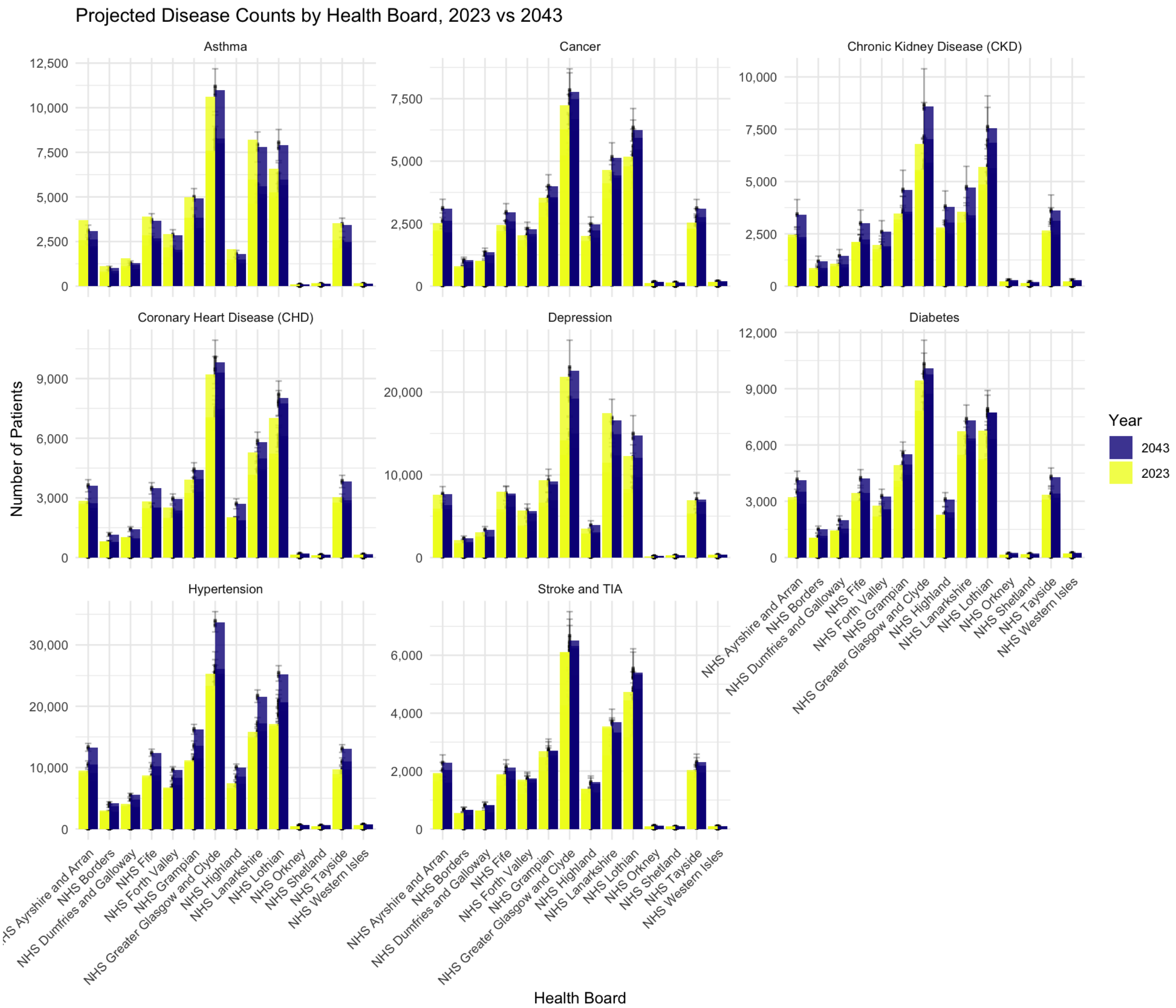

Figure 10: Disease counts by health board in 2023 and 2043, side by side.

6.4 Future Work

Future work could address several of these limitations. Expanding prevalence series to cover more years and ensure full practice reporting would improve model stability. Developing directly observed subnational projections would reduce reliance on proportional scaling. Integrating richer workforce measures—such as GP working hours, skill mix, and practice organisation—would allow supply-side pressures to be modelled more realistically. Finally, incorporating dynamic epidemiological trends and risk factors would enable projections to move beyond demographic extrapolation toward fuller representations of population health change.

7 Conclusion

This study presents a novel set of forecasts of GP disease burden in Scotland, combining recently released practice-level prevalence data with updated, post-Brexit demographic projections. The results point to substantial growth in demand, particularly for cardiovascular, metabolic, and mental health conditions, with most of the increase attributable to population ageing. While the analysis is constrained by fragmented data sources, short prevalence series, and simplifying assumptions, it nonetheless provides a consistent and transparent framework for anticipating future pressures on GP services.

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